PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 2 • Chapter OL-T23 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

5 classified items • 16 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T23 • Expertise Q16+ • 5 items • 16 marks

Chapter	Items	Marks	Starts
01. OL-T23 Algebraic Proof	5	16	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Algebraic Proof

Unit 2 • Expertise • 16 marks • 5 item(s)

January 2022 | 4MA1-2H Q18 | 3 marks
Fixed remainder modulo a divisor

January 2020 | 4MA1-2H Q17 | 3 marks
Consecutive integer parity identity

June 2021 | 4MA1-1H Q17 | 3 marks
Consecutive even/odd sequence proof

November 2021 | 4MA1-1H Q19 | 3 marks
Derive a general geometric algebra formula

January 2023 | 2H Q17 | 4 marks
Product or sum of arbitrary odd/even numbers

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q18 **3 marks**

18 Prove that when the sum of the squares of any two consecutive odd numbers is divided by 8, the remainder is always 2
Show clear algebraic working.

WORKING SPACE

January 2020 | 4MA1-2H | Question 17 | 3 marks

Chapter OL-T23: Algebraic Proof

17 Prove that the difference between two consecutive square numbers is always an odd number.
Show clear algebraic working.

YOUR WORKING

June 2021 | 4MA1-1H | Question 17 | 3 marks

Chapter OL-T23: Algebraic Proof

17 Using algebra, prove that, given any 3 consecutive even numbers, the difference between the square of the largest number and the square of the smallest number is always 8 times the middle number.

YOUR WORKING

19 *ABCED* is a five-sided shape.

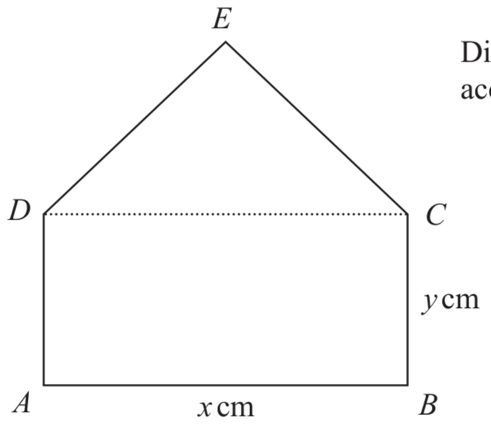

Diagram **NOT** accurately drawn

ABCD is a rectangle.
CED is an equilateral triangle.

$AB = x$ cm $\quad BC = y$ cm

The perimeter of *ABCED* is 100 cm.
The area of *ABCED* is R cm^2

(a) Show that $R = \frac{x}{4}\left(200 - \left[6 - \sqrt{3}\right]x\right)$

(3)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$x =$ ....

(2)

(ii) Explain why the maximum value of R is given by this value of x.

....

....

....

(1)

YOUR WORKING

January 2023 | 2H | Question 17 | 4 marks

Chapter OL-T23: Algebraic Proof

17 Given that $8\sqrt{m} + \sqrt{49m} - \sqrt{9m} = k\sqrt{m}$

where k is an integer and m is a prime number,

(a) work out the value of k

$k =$ ....

(1)

(b) Show that $\dfrac{5 - \sqrt{18}}{1 - \sqrt{2}}$ can be written in the form $a + b\sqrt{2}$

where a and b are integers.

Show each stage of your working clearly.

(3)

(Total for Question 17 is 4 marks)

YOUR WORKING